

Chapter 2 Fatigue

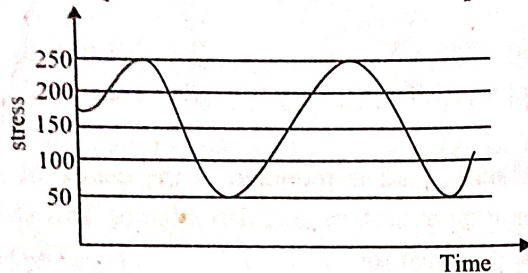
One Mark Questions

01. Stress concentration in a machine component of a ductile material is not so harmful as it is in a brittle material because (GATE-ME-89)
 - (a) in ductile material local yielding may distribute stress concentration
 - (b) ductile material has larger Young's modulus.
 - (c) Poisson's ratio is larger in ductile materials.
 - (d) Modulus of rigidity is larger in ductile materials.
02. The process of shot peening increases the fatigue life of steel springs mainly because it results in (GATE-ME-90)
 - (a) surface hardening
 - (b) increased stiffness of the material
 - (c) structural changes in the material
 - (d) residual compression at the surface
03. In a shaft with a transverse hole, as the hole to the shaft diameter ratio _____ (increases /decreases), the torsional stress concentration factor _____ (increases/decreases) (GATE-ME-91)
04. Fatigue strength of rod subjected to cyclic axial force is less than that of a rotating beam of the same dimensions subjected to steady lateral force because (GATE-ME-92)
 - (a) axial stiffness is less than bending stiffness
 - (b) of absence of centrifugal effects in the rod
 - (c) the number of discontinuities vulnerable to fatigue are more in the rod
 - (d) at a particular time the rod has only one type of stress whereas the beam has both the tensile and compressive stresses
05. The yield strength of a steel shaft is twice its endurance limit. Which of the following torque fluctuations represent the most critical situation according to Soderberg criterion? (GATE-ME-93)

- (a) $-T$ to $+T$
- (b) $-T/2$ to $+T$
- (c) 0 to $+T$
- (d) $+T/2$ to $+T$

06. A static load is mounted at the centre of a shaft rotating at uniform angular velocity. This shaft will be designed for (GATE-ME-02)
 - (a) the maximum compressive stress (static)
 - (b) the maximum tensile stress (static)
 - (c) the maximum bending moment (static)
 - (d) fatigue loading
07. In terms of theoretical stress concentration factor (k_t) and fatigue stress concentration factor (k_f), the notch sensitivity 'q' is expressed as (GATE-ME-03)
 - (a) $(k_f - 1)/(k_t - 1)$
 - (b) $(k_f - 1)/(k_t + 1)$
 - (c) $(k_t - 1)/(k_f - 1)$
 - (d) $(k_t + 1)/(k_f + 1)$
08. The S-N curve for steel becomes asymptotic nearly at (GATE-ME-03)
 - (a) 10^3 cycles
 - (b) 10^4 cycles
 - (c) 10^6 cycles
 - (d) 10^9 cycles
09. In a structure subjected to fatigue loading, the minimum and maximum stresses developed in a cycle are 200 MPa and 400 MPa respectively. The value of stress amplitude (in MPa) is _____ (GATE -14-SET-2)
10. A rotating steel shaft is supported at the ends. It is subjected to a point load at the center. The maximum bending stress developed is 100 MPa. If the yield, ultimate and corrected endurance strength of the shaft material are 300 MPa, 500 MPa and 200 MPa, respectively, then the factor of safety for the shaft is _____ (GATE-14-SET-3)
11. Which one of the following is the most conservative fatigue failure criterion? (GATE -15 -Set 1)
 - (a) Soderberg
 - (b) Modified Goodman
 - (c) ASME Elliptic
 - (d) Gerber

12. For the given fluctuating fatigue load, the value of stress amplitude and stress ratio are respectively.



(GATE -15 -Set 3)

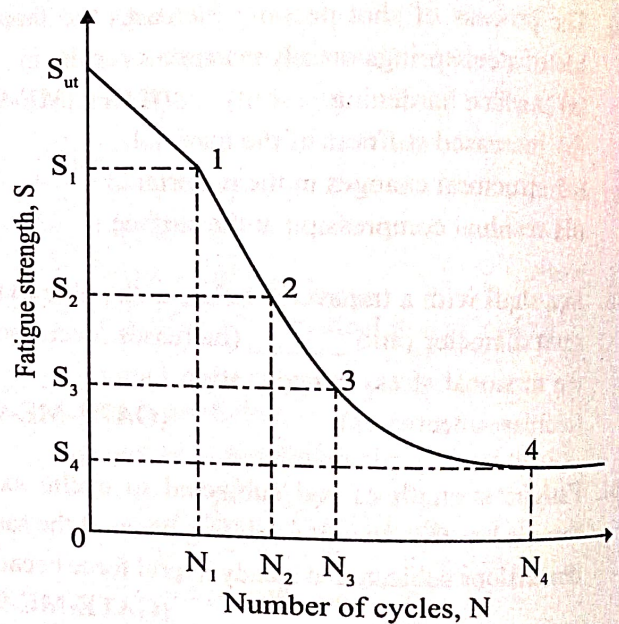
- (a) 100 MPa and 5 (b) 250 MPa and 5
(c) 100 MPa and 0.20 (d) 250 MPa and 0.20
13. A machine component made of a ductile material is subjected to a variable loading with $\sigma_{\min} = -50$ MPa and $\sigma_{\max} = 50$ MPa. If the corrected endurance limit and the yield strength for the material are $\sigma'_e = 100$ MPa and $\sigma_y = 300$ MPa, respectively, the factor of safety is ____ (GATE - 17 - SET - 2)
14. Fatigue life of a material for a fully reversed loading condition is estimated from

$$\sigma_a = 1100 N^{-0.15},$$
 where σ_a is the stress amplitude in MPa and N is the failure life in cycles. The maximum allowable stress amplitude (in MPa) for a life of 1×10^5 cycles under the same loading condition is ____ (correct to two decimal places). (GATE-18-SET-2)
15. During a high cycle fatigue test, a metallic specimen is subjected to cyclic loading with a mean stress of +140 MPa, and a minimum stress of -70 MPa. The R-ratio (minimum stress to maximum stress) for this cyclic loading is ____ (round off to one decimal place) (GATE - 19 - SET - 1)
16. Endurance limit of a beam subjected to pure bending decreases with (GATE - 19 - SET - 2)
 (a) increase in the surface roughness and increase in the size of the beam
 (b) increase in the surface roughness and decrease in the size of the beam

- (c) decrease in the surface roughness and decrease in the size of the beam
 (d) decrease in the surface roughness and increase in the size of the beam

17. A machine member is subjected to fluctuating stress $\sigma = \sigma_0 \cos(8\pi t)$. The endurance limit of the material is 350 MPa. If the factor of safety used in the design is 3.5 then the maximum allowable value of σ_0 is ____ MPa (round off to 2 decimal places). (GATE-20-SET-2)

18. The S-N curve from a fatigue test for steel is shown. Which one of the options gives the endurance limit? (GATE_ME-23)



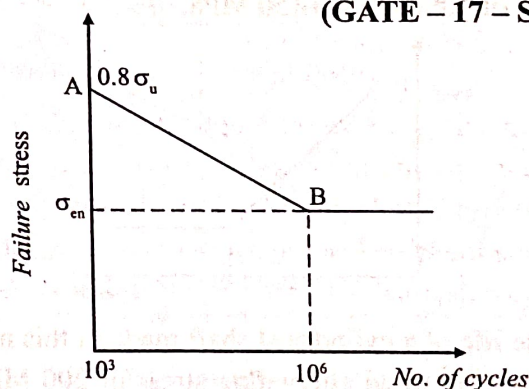
- (a) S_{ut} (b) S_2
(c) S_3 (d) S_4

Two Marks Questions

01. A ductile material having an endurance limit of 196 N/mm^2 and the yield point at 294 N/mm^2 is stressed under variable load. The maximum and minimum stresses are 147 N/mm^2 and 49 N/mm^2 . The fatigue stress concentration factor is 1.32. The available factor of safety for this loading is (GATE-ME-87)
(a) 3.0 (b) 1.5 (c) 1.33 (d) 4.0
02. A cylindrical shaft is subjected to an alternating stress of 100 MPa . Fatigue strength to sustain 1000 cycles is 490 MPa . If the corrected endurance strength is 70 MPa , estimated shaft life will be (GATE-ME-06)
(a) 1071 cycles (b) 15000 cycles
(c) 281914 cycles (d) 928643 cycles
03. A thin spherical pressure vessel of 200 mm diameter and 1 mm thickness is subjected to an internal pressure varying from 4 to 8 MPa . Assume that the yield, ultimate and endurance strength of material are 600 , 800 , 400 MPa respectively. The factor of safety as per Goodman's relation is (GATE-ME-07)
(a) 2.0 (b) 1.6 (c) 1.4 (d) 1.2
04. A forged steel link with uniform diameter of 30 mm at the centre is subjected to an axial force that varies from 40 kN in compression to 160 kN in tension. The tensile (S_u), yield (S_y) and corrected endurance (S_e) strengths of steel material are 600 MPa , 420 MPa and 240 MPa respectively. The factor of safety against fatigue endurance as per Soderberg's criterion is (GATE-ME-09)
(a) 1.26 (b) 1.37 (c) 1.45 (d) 2.00
05. A bar is subjected to fluctuating tensile load from 20 kN to 100 kN . The material has yield strength of 240 MPa and endurance limit in reversed bending is 160 MPa . According to the Soderberg's principle,

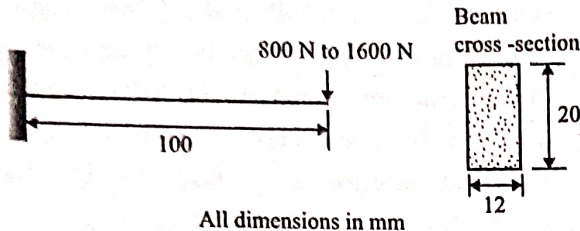
the area of cross-section in mm^2 of the bar for a factor of safety of 2 is (GATE-ME-13)
(a) 400 (b) 600 (c) 750 (d) 1000

06. In a structural member under fatigue loading, the minimum and maximum stresses developed at the critical point are 50 MPa and 150 MPa , respectively. The endurance, yield, and the ultimate strengths of the material are 200 MPa , 300 MPa , and 400 MPa , respectively. The factor of safety using modified Goodman criterion is (GATE - 16 - SET - 2)
(a) $\frac{3}{2}$ (b) $\frac{8}{5}$ (c) $\frac{12}{7}$ (d) 2
07. A machine element has an ultimate strength (σ_u) of 600 N/mm^2 , and endurance limit (σ_{en}) of 250 N/mm^2 . The fatigue curve for the element on a log-log plot is shown below. If the element is to be designed for a finite life of 10000 cycles, the maximum amplitude of a completely reversed operating stress is _____ N/mm^2 . (GATE - 17 - SET - 1)



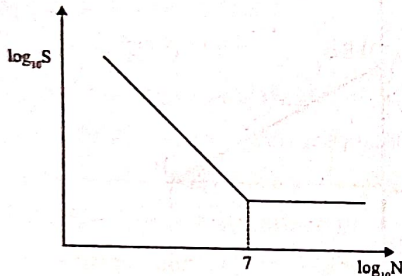
08. A bar is subjected to combination of a steady load of 60 kN and a load fluctuating between -10 kN and 90 kN . The corrected endurance limit of the bar is 150 MPa , the yield strength of the material is 480 MPa and the ultimate strength of the material is 600 MPa . The bar cross-section is square with side a . If the factor of safety is 2, the value of a (in mm), according to the modified Goodman's criterion, is _____ (correct to two decimal places). (GATE - 18 - SET - 2)

09. A machine part in the form of cantilever beam is subjected to fluctuating load as shown in the figure. The load varies from 800 N to 1600 N. The modified endurance, yield and ultimate strengths of the material are 200 MPa, 500 MPa and 600 MPa, respectively.



The factor of safety of the beam using modified Goodman criterion is _____ (round off to one decimal place). (GATE-21_SET-1)

10. The figure shows the relationship between fatigue strength (S) and fatigue life (N) of a material. The fatigue strength of the material for a life of 1000 cycles is 450 MPa, while its fatigue strength for a life of 10^6 cycles is 150 MPa.



The life of a cylindrical shaft made of this material subjected to an alternating stress of 200 MPa will then be _____ cycles (round off to the nearest integer) (GATE-21_SET-2)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (a)	02. (d)	03. (*)	04. (c)	05. (a)
06. (d)	07. (a)	08. (c)	09. 100 MPa	
10. Ans. 2	11. (a)	12. (c)	13. 2	14. 195.61
15. -0.2	16. (a)	17. 100	18. (d)	

TWO MARKS QUESTIONS

01. (b)	02. (c)	03. (b)	04. (a)	05. (d)
06. (d)	07. 385.42	08. 31.62		
09. 2	10. 163840.58			

One Mark Solutions

01. Ans: (a)

Sol: The effect of stress concentration factor is not considered harmful for ductile material in static loading because when the stress near the discontinuity reaches the yield point, there is plastic deformation, resulting in the redistribution of stresses. Thus there is no noticeable damage to the part as a whole.

02. Ans: (d)

Sol: Shot peening is a cold working process used to produce a compressive residual stress layer and modify mechanical properties.

03. Ans: Increases, Decreases

Sol: Stress concentration factor is due to sudden change of cross-sectional area. If cross-sectional area changes more suddenly, then stress concentration factor will be more and vice versa i.e. small holes lead to more stress concentration factor.

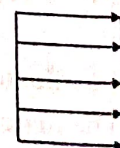
- Hence as the ratio of hole to shaft diameters increases which means hole diameter is more, i.e. change of cross-section is not more, so the corresponding stress concentration factor will be less.

04. Ans: (c)

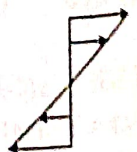
Sol:



Cross-section



Axial stress variation



Bending stress variation

- From above stress distribution diagrams, it is clear that for axial loading, entire cross section is under maximum stress. But for bending loading, only extreme fibres are under maximum stress. The cross-section or fibres near neutral axis are under low stress.
- Hence probability of having a micro-crack in maximum stress region in axial loading is more than that of in bending loading.
- In other words, axially loaded component can start failing from any point while bending component can start failing from extreme regions only.
- Therefore, endurance limit in axial loading is lower than the rotating beam element.
- In rotating beam test, component is subjected to reversed bending.

05. Ans: (a)

Sol: Soderberg's equation,

$$\frac{\tau_a}{S_{se}} + \frac{\tau_m}{S_{sy}} = \frac{1}{FS}$$

$$\frac{\tau_a}{S_{se}} + \frac{\tau_m}{2 \times S_{se}} = \frac{1}{FS}$$

$$\tau_a + \frac{\tau_m}{2} = \frac{S_{se}}{FS}$$

$$\Rightarrow \tau_a = \frac{16T_a}{\pi d^3} \Rightarrow \tau_a \propto T_a$$

$$\tau_m = \frac{16T_m}{\pi d^3} \Rightarrow \tau_m \propto T_m$$

Options	T_a	T_m	$T_a + \frac{T_m}{2}$
a	T	0	T
b	$\frac{3T}{4}$	$\frac{T}{4}$	$\frac{7}{8}T$
c	$\frac{T}{2}$	$\frac{T}{2}$	$\frac{3}{4}T$
d	$\frac{T}{4}$	$\frac{3T}{4}$	$\frac{5}{8}T$

The stress for option (a) is more and hence critical.

06. Ans: (d)

Sol: A static load is mounted at the centre of a shaft rotating at uniform angular velocity, experiences fluctuating stress which will be designed for fatigue loading.

07. Ans: (a)

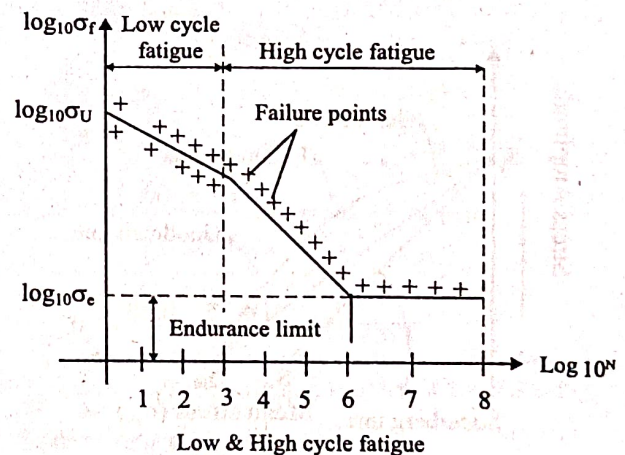
Sol: $q = \frac{\text{increase of actual stress over nominal stress}}{\text{increase of theoretical stress over nominal stress}}$

$$q = \frac{\sigma k_t - \sigma}{\sigma k_t - \sigma}$$

$$\Rightarrow q = \frac{k_t - 1}{k_t - 1} (0 < q < 1)$$

08. Ans: (c)

Sol: S-N curve for ferrous material like steel is represented below.



Low and High cycle fatigue [stresses amplitude (σ_f) versus the number of stress cycles (N)].

For Ferrous metals like steel, S-N curve becomes asymptotic at 10^6 cycles, represents endurance limit.

09. Ans: 100 MPa

Sol: $\sigma_{min} = 200$ MPa

$\sigma_{max} = 400$ MPa

$$\text{Stress amplitude } \sigma_a = \frac{400 - 200}{2} = 100 \text{ MPa}$$

10. Ans: 2

Sol: $\sigma_{\max} = 100 \text{ MPa}$

Bending stress is a completely reversible stress and for bending stress mean stress,

$$\sigma_m = 0$$

$\Rightarrow$ Stress amplitude ' σ_a ' = 100 MPa

$$S_{yt} = 300 \text{ MPa},$$

$$S_{ut} = 500 \text{ MPa}$$

$$S_e = 200 \text{ MPa},$$

$$F.S = ?$$

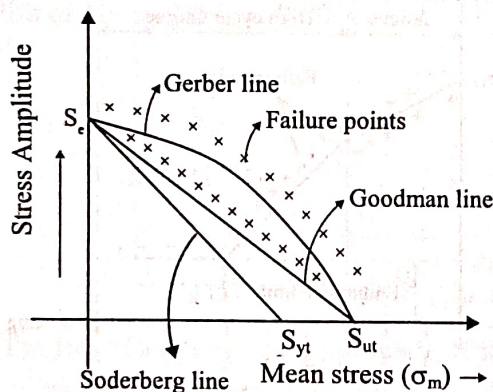
From Soderberg's equation $\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{1}{F.S}$

$$\frac{100}{200} = \frac{1}{F.S} \Rightarrow F.S = 2$$

11. Ans: (a)

Sol: According Soderberg equation,

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{1}{\text{Factor of Safety}}$$



12. Ans: (c)

Sol: $\sigma_{\min} = 50 \text{ MPa}$

$$\sigma_{\max} = 250 \text{ MPa}$$

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{250 - 50}{2} = 100 \text{ MPa}$$

$$\text{Stress ratio, } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{50}{250} = \frac{1}{5} = 0.20$$

13. Ans: 2

Sol: $\sigma_{\min} = -50 \text{ MPa},$

$$s_e = 100 \text{ MPa},$$

$$\sigma_{\max} = 50 \text{ MPa},$$

$$S_{yt} = 300 \text{ MPa}$$

The stress is completely reversed stress and mean stress, $\sigma_m = 0$ and stress amplitude. $\sigma_a = 50 \text{ MPa}.$

$$\sigma_a = \frac{S_e}{F.S}$$

$$50 = \frac{100}{F.S} \Rightarrow F.S = 2$$

14. Ans: 195.61

Sol: Stress amplitude (variable stress) = $\sigma_a = 1100 \text{ N}^{-0.15}$

$$N = 1 \times 10^5 \text{ cycle}$$

$$\therefore \sigma_a = 1100 \times (1 \times 10^5)^{-0.15} = 195.61 \text{ MPa}$$

15. Ans: -0.2

Sol: $\sigma_{\min} = -70 \text{ MPa}$

$$\sigma_{\text{mean}} = 140 \text{ MPa}$$

$$\Rightarrow \sigma_{\text{mean}} = \frac{\sigma_{\min} + \sigma_{\max}}{2} = 140$$

$$\Rightarrow \frac{-70 + \sigma_{\max}}{2} = 140$$

$$\Rightarrow \sigma_{\max} = 350 \text{ MPa}$$

$$\text{The R-ratio, } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{-70}{350} = -0.2$$

16. Ans: (a)

Sol: Endurance limit decreases with increase in surface roughness and with increase in size of the beam.

17. Ans: 100

Sol: Given: $\sigma = \sigma_0 \cdot \cos(8\pi t)$

$\cos \theta$ varies from 1 to -1

$\therefore \sigma$ varies from σ_0 to $-\sigma_0$

$$\sigma_{\max} = \sigma_0 \quad \& \quad \sigma_{\min} = -\sigma_0$$

Here, the loading is reversed.

For reversed loading design condition is,

$$\text{Amplitude stress } \sigma_a = \frac{\sigma_e}{\text{fos}}$$

Where, σ_e = Corrected endurance strength

$$\therefore \sigma_0 = \frac{350}{3.5}$$

$$\sigma_0 = 100 \text{ MPa}$$

18. Ans: (d)

Sol: Endurance strength: It is the maximum amplitude stress that the component can withstand a minimum life of 10^6 cycles without fatigue failure.

∴ From graph S_e represents endurance strength.

Two Marks Solutions

01. Ans: (b)

$$\text{Sol: } \sigma_m = \frac{147 + 42}{2} = 98 \text{ MPa}$$

$$\sigma_a = \frac{147 - 42}{2} = 49 \text{ MPa}$$

$$S_e = \frac{196}{1.32} = 148.48 \text{ MPa}$$

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{1}{F.S}$$

$$\Rightarrow \frac{49}{148.48} + \frac{98}{294} = \frac{1}{F.S}$$

$$\Rightarrow F.S = 1.50$$

02. Ans: (c)

Sol: Basquin's equation, $\sigma_a = AN^B$

$$490 = A \times 10^{3B} \dots\dots\dots (1)$$

$$70 = A \times 10^{6B} \dots\dots\dots (2)$$

Solving 1 and 2,

$$A = \frac{70}{10^{6 \times (-0.2817)}} = 3430, \quad B = -0.2817$$

$$N = \left(\frac{\sigma_a}{A} \right)^{1/B} = 281913.59 \text{ cycles}$$

03. Ans: (b)

Sol: $D = 200 \text{ mm}, t = 1 \text{ mm}, P_{\min} = 4 \text{ MPa}$

$$P_{\max} = 8 \text{ MPa}, \quad S_{yt} = 600 \text{ MPa},$$

$$S_{ut} = 800 \text{ MPa}, \quad S_e = 400 \text{ MPa}$$

$$\sigma_h = \sigma_{1\max} = \frac{PD}{4t} = \frac{8 \times 200}{4 \times 1} = 400 \text{ MPa}$$

$$\sigma_{1\min} = \frac{PD}{4t} = \frac{4 \times 200}{4 \times 1} = 200 \text{ MPa}$$

$$\sigma_1 = \sigma_{2\max} = 400 \text{ MPa}$$

$$\sigma_{2\min} = 200 \text{ MPa}$$

$$\sigma_{1m} = \frac{\sigma_{1\max} + \sigma_{1\min}}{2} = \frac{400 + 200}{2} = 300 \text{ MPa}$$

$$\sigma_{1a} = \frac{\sigma_{1\max} - \sigma_{1\min}}{2} = \frac{400 - 200}{2} = 100 \text{ MPa}$$

$$\sigma_{2m} = 300 \text{ MPa},$$

$$\sigma_{2a} = 100 \text{ MPa}$$

According to distortion energy theory,

$$\sigma_{mes} = \sqrt{\sigma_{1m}^2 + \sigma_{2m}^2 - \sigma_{1m}\sigma_{2m}} = 300 \text{ MPa}$$

$$\sigma_{aes} = \sqrt{\sigma_{1a}^2 + \sigma_{2a}^2 - \sigma_{1a}\sigma_{2a}} = 100 \text{ MPa}$$

Goodman's equation

$$\frac{\sigma_{aes}}{S_e} + \frac{\sigma_{mes}}{S_{ut}} = \frac{1}{F.S}$$

$$\Rightarrow \frac{100}{400} + \frac{300}{800} = \frac{1}{F.S} \Rightarrow F.S = 1.6$$

04. Ans: (a)

$$\text{Sol: } P_m = \frac{160 - 40}{2} = 60 \text{ kN}$$

$$\Rightarrow \sigma_m = \frac{P_m}{A}$$

$$= \frac{60 \times 10^3 \text{ N}}{\frac{\pi}{4} (30)^2 \text{ mm}^2} = 84.88 \text{ N/mm}^2$$

$$P_a = \frac{160 - (-40)}{2} = 100 \text{ kN}$$

$$\Rightarrow \sigma_a = 141.47 \text{ N/mm}^2$$

Soderberg's Equation:

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{1}{F.S}$$

$$\Rightarrow \frac{141.47}{240} + \frac{84.88}{420} = \frac{1}{F.S}$$

$$\Rightarrow F.S = 1.26$$

05. Ans: (d)

$$\text{Sol: } P_{\max} = 100 \text{ kN}, \quad P_{\min} = 20 \text{ kN},$$

$$S_{yt} = 240 \text{ MPa}, \quad S_e = 160 \text{ MPa},$$

$$A = ?, \quad F.S = 2$$

$$\sigma_{\max} = \frac{100 \times 10^3 \text{ N}}{A \text{ mm}^2}$$

$$\sigma_{\min} = \frac{20 \times 10^3 \text{ N}}{A \text{ mm}^2}$$

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{80 \times 10^3}{2A} = \frac{40 \times 10^3}{A} \text{ N/mm}^2$$

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2} = \frac{60 \times 10^3}{A} \text{ N/mm}^2$$

Soderberg's equation $\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{1}{FS}$

$$\frac{10^3}{A} \left[\frac{40}{60} + \frac{60}{240} \right] = \frac{1}{2}$$

$$\frac{10^3}{A} \left[\frac{1}{4} + \frac{1}{4} \right] = \frac{1}{2}$$

$$\frac{10^3}{A} \left[\frac{1}{2} \right] = \frac{1}{2} \Rightarrow A = 1000 \text{ mm}^2$$

06. Ans: (d)

Sol: $\sigma_{\max} = 150 \text{ MPa}$, $\sigma_{\min} = 50 \text{ MPa}$
 $S_e = 200 \text{ MPa}$, $S_{yt} = 300 \text{ MPa}$,
 $S_{ut} = 400 \text{ MPa}$

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{150 - 50}{2} = 50 \text{ MPa}$$

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2} = \frac{150 + 50}{2} = 100 \text{ MPa}$$

Modified Goodman's Theory:

First check the slope of $\frac{\sigma_m}{S_e}$ & $\frac{H}{K}$,

If, $\frac{\sigma_a}{\sigma_m} \leq \frac{H}{K}$, then use Goodman's theory

If, $\frac{\sigma_a}{\sigma_m} > \frac{H}{K}$, then use Langer's theory'

Langer theory: $\frac{H}{S_{yt}} + \frac{K}{S_{yt}} = 1$

$$H + K = S_{yt}$$

$$H + K = 300 \dots\dots\dots (i)$$

Goodman theory:

$$\frac{H}{S_{ut}} + \frac{K}{S_e} = 1$$

$$\frac{H}{400} + \frac{K}{200} = 1$$

$$\frac{H}{400} + \frac{K \times 2}{200 \times 2} = 1$$

$$H + K + K = 400 \dots\dots\dots (ii)$$

from equation 1 & 2 :

$$300 + K = 400 \Rightarrow K = 100 \text{ \& } H = 200$$

So, $\frac{K}{H} = \frac{100}{200} = \frac{1}{2}$

and $\frac{\sigma_a}{\sigma_m} = \frac{50}{100} = \frac{1}{2}$

Here $\frac{K}{H} = \frac{\sigma_a}{\sigma_m}$ So, we can use either Langer's theory or Goodman theory.

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{FS}$$

$$\frac{50}{200} + \frac{100}{400} = \frac{1}{FS} \Rightarrow F.S = 2$$

07. Ans: (385.42)

Sol: $S_{ut} = 600 \text{ MPa}$, $S_e = 250 \text{ MPa}$,
 $L = 10000 \text{ cycles}$, $S_f = ?$

Basquin's equation,

$$A = s_f L^B$$

$$A = (0.8 \times 600) \times 10^{3B} \dots\dots\dots (1)$$

$$A = 250 \times 10^{6B} \dots\dots\dots (2)$$

Dividing equation (2) with equation (1)

$$1 = 0.520 \times 10^{3B}$$

$$B = 0.094$$

$$\therefore A = 250 \times 10^{6 \times 0.094} = 916.09$$

$$\therefore 916.09 = s_f L^{0.094}$$

$$916.09 = s_f \times 10^4 \times 0.094$$

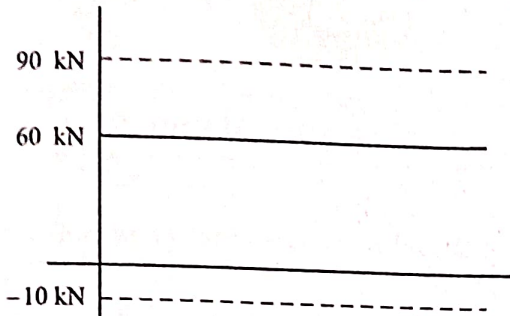
$$s_f = 385.42 \text{ MPa}$$

08. Ans: 31.62

Sol: $P_m = 60 \text{ kN}$; $P_v = -10 \text{ kN to } 90 \text{ kN}$
 $\sigma_e = 150 \text{ MPa}$; $\sigma_y = 480 \text{ MPa}$
 $\sigma_u = 600 \text{ MPa}$

Cross section = square = a^2

$$FS = 2$$



As per Goodman's equation

$$\frac{\sigma_m}{\sigma_n} + \frac{\sigma_v}{\sigma_e} = \frac{1}{FS}$$

$$\therefore \text{Mean load} = 60 \text{ kN} + \frac{-10 + 90}{2}$$

$$P_m = 100 \text{ kN}$$

$$P_v = \frac{90 - (-10)}{2} = 50 \text{ kN}$$

$$\sigma_m = \frac{P_m}{a^2} = \frac{100 \times 10^3}{a^2}$$

$$\sigma_v = \frac{P_v}{a^2} = \frac{50 \times 10^3}{a^2}$$

Goodmans's equation

$$\frac{100 \times 10^3}{a^2 \times 600} + \frac{50 \times 10^3}{a^2 \times 150} = \frac{1}{2}$$

$$\therefore a^2 = 1000 \text{ mm}^2$$

$$\therefore a = 31.62 \text{ mm}$$

09. Ans: (2)

Sol:

$$(i) P_m = \frac{P_{\max} + P_{\min}}{2} = \frac{1600 + 800}{2} = 1200 \text{ N}$$

$$P_a = \frac{P_{\max} - P_{\min}}{2} = \frac{1600 - 800}{2} = 400 \text{ N}$$

(ii) Bending stress,

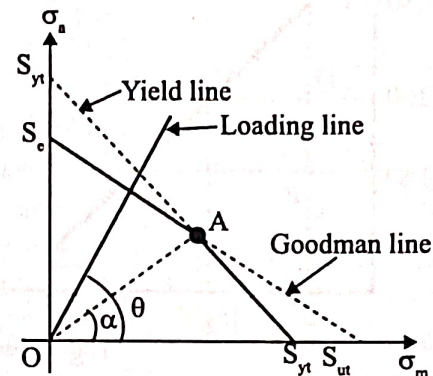
$$\sigma = \frac{M y}{I} = \frac{P \times L \times \frac{t}{2}}{\frac{bt^3}{12}} = \frac{6P \times L}{bt^2}$$

$$\sigma_m = \frac{6 \times P_m \times L}{b \times t^2} = \frac{6 \times 1200 \times 100}{12 \times 20^2} = 150 \text{ MPa}$$

$$\sigma_a = \frac{6 \times P_a \times L}{b \times t^2} = 50 \text{ MPa}$$

$$(iii) \text{ Loading line, } \tan \theta = \frac{\sigma_a}{\sigma_m} = \frac{50}{150} = \frac{1}{3} \text{ ---- (i)}$$

(iv) Modified Goodman diagram,



$$\begin{aligned} \text{Yield line} &\Rightarrow \sigma_m + \sigma_a = S_y \\ &\Rightarrow \sigma_m + \sigma_a = 500 \text{ ---- (ii)} \end{aligned}$$

Goodman line

$$\frac{\sigma_m}{S_{ut}} + \frac{\sigma_a}{S_e} = 1$$

$$\frac{\sigma_m}{600} + \frac{\sigma_a}{200} = 1$$

$$\Rightarrow \sigma_m + 3\sigma_a = 600 \text{ ---- (iii)}$$

On solving eq.(ii) and (iii)

$$\Rightarrow \sigma_a = 50 \text{ MPa}$$

$$\Rightarrow \sigma_m = 450 \text{ MPa}$$

$$\tan \alpha = \frac{\sigma_a}{\sigma_m} = \frac{50}{450} = \frac{1}{9} \text{ ---- (iv)}$$

From (i) and (iv)

$$\tan \theta > \tan \alpha$$

$$\therefore \theta > \alpha$$

Design line is Goodman line,

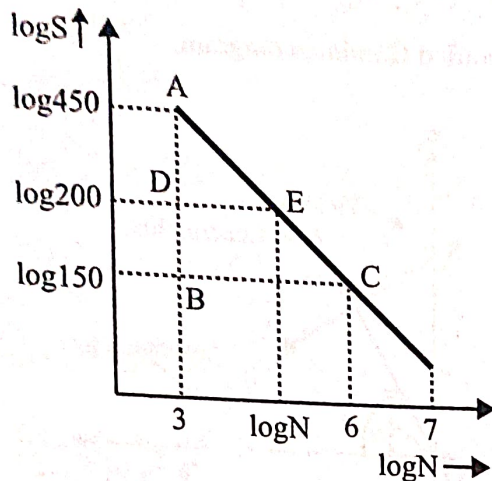
$$\frac{\sigma_m}{S_{ut}} + \frac{\sigma_a}{S_e} = \frac{1}{FOS}$$

$$\frac{150}{600} + \frac{50}{200} = \frac{1}{FOS}$$

$$\therefore FOS = 2$$

10. Ans: (163840.58)

Sol:



Comparing similar triangles,
 $\triangle ABC$ and $\triangle ADE$

$$\frac{AB}{BC} = \frac{AD}{ED}$$

$$\frac{\log 450 - \log 150}{6 - 3} = \frac{\log 450 - \log 200}{\log N - 3}$$

$$\therefore N = 163840.58 \text{ cycles}$$

Chapter

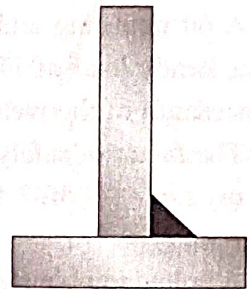
3

Bolted, Riveted and Welded joints

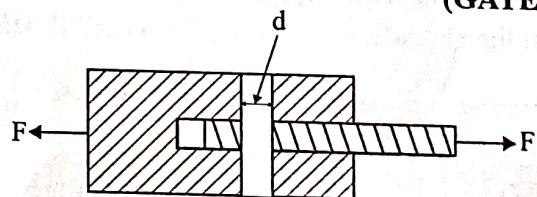
One Mark Questions

01. Weldments in fabricated steel beams are designed for (GATE-ME-87)
 - (a) bending stresses at the flange
 - (b) shear stresses in transverse plane
 - (c) combination of bending and shear
 - (d) none of these because in fabricated beams welds do not get stressed.
02. If the ratio of the diameter of rivet hole to the pitch of rivets is 0.25, then the tearing efficiency of the joint is (GATE-ME-96)
 - (a) 0.50
 - (b) 0.75
 - (c) 0.25
 - (d) 0.87
03. The bolts in a rigid flanged coupling connecting two shafts transmitting power are subjected to (GATE-ME-96)
 - (a) shear force and bending moment
 - (b) axial force
 - (c) torsion
 - (d) torsion and bending moment
04. Bolts in the flanged end of pressure vessel are usually pre-tensioned. Indicate which of the following statements is not true (GATE-ME-98)
 - (a) Pre-tensioning helps to seal the pressure vessel
 - (b) Pre-tensioning increase the fatigue life of the bolts
 - (c) Pre-tensioning reduces the maximum tensile stress in the bolts
 - (d) Pre-tensioning helps to reduce the effect of pressure pulsations in the pressure vessels
05. The type of weld represented by the shaded region in the figure is (GATE-18-SET-1)

- (a) groove
- (b) spot
- (c) fillet
- (d) plug



06. Pre-tensioning of a bolted joint is used to (GATE - 18 - SET - 2)
 - (a) strain harden the bolt head
 - (b) decrease stiffness of the bolted joint
 - (c) increase stiffness of the bolted joint
 - (d) prevent yielding of the thread root
07. The figure shows two bodies connected through a riveted joint with one rivet. The diameter of the rivet is d (in m). The joint transmits a load of F (in N) whose line of action is perpendicular to and intersects the vertical axis of the rivet. Neglect any effect of bending of the rivet. If the allowable shear stress for the material of the rivet is τ N/m², the diameter of the rivet required to prevent failure in shear is (GATE-PI-20)

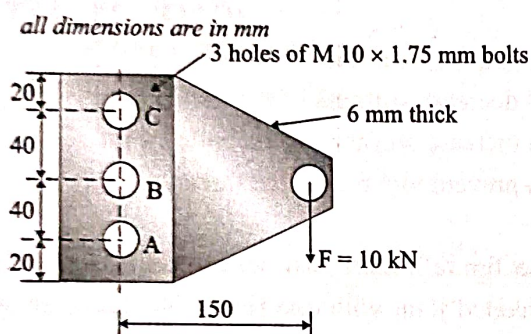


- (a) $\sqrt{\frac{F}{\pi\tau}}$
 - (b) $\sqrt{\frac{8F}{\pi\tau}}$
 - (c) $\sqrt{\frac{2F}{\pi\tau}}$
 - (d) $\sqrt{\frac{4F}{\pi\tau}}$
08. A M30 bolt needs to be subjected to pretension $F_t = 350$ kN. If the torque coefficient K of the bolt is 0.2, then the torque in Nm needed to achieve this pretension is _____ [in integer] (GATE-PI-21)

Two Marks Questions

01. A 60 mm long and 6 mm thick fillet weld carries a steady load of 15 kN along the weld. The shear strength of the weld material is equal to 200 MPa. The factor of safety is (GATE-ME-06)
(a) 2.4 (b) 3.4 (c) 4.8 (d) 6.8

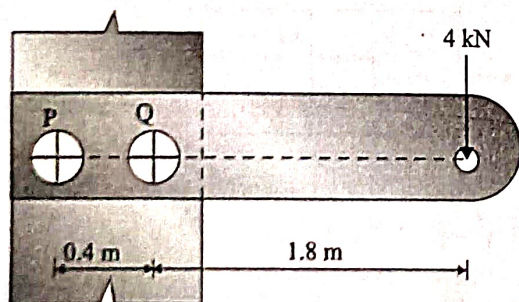
02. A bolted joint is shown below. The maximum shear stress, in MPa, in bolts A and B respectively are (GATE-ME-07)



- (a) 242.6, 42.5 (b) 42.5, 242.6
(c) 42.5, 42.5 (d) 242.6, 242.6

Statement for Linked Answer Q.Nos. 03 & 04

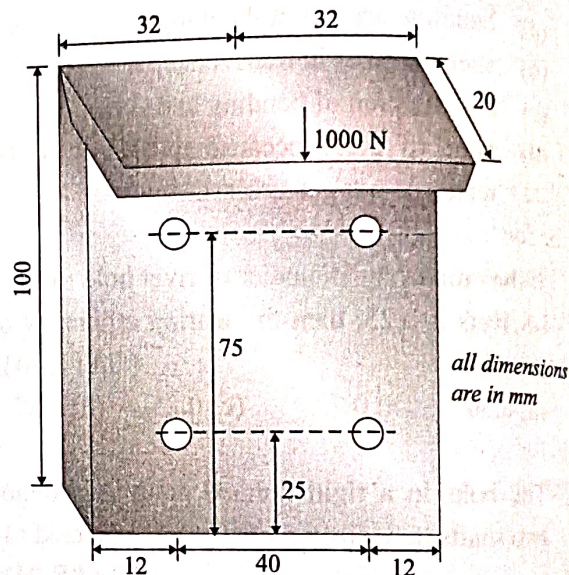
A steel bar of 10 mm × 50 mm is cantilevered with two M12 bolts (P and Q) to support a static load of 4 kN as shown in figure aside. (GATE-ME-08)



03. The primary and secondary shear loads on rivet p, respectively are
(a) 2 kN, 20 kN (b) 20 kN, 2 kN
(c) 20 kN, 0 kN (d) 0 kN, 20 kN

04. The resultant shear stress on rivet P is closest to
(a) 132 MPa (b) 159 MPa
(c) 178 MPa (d) 195 MPa

05. A bracket shown in fig. is rigidly mounted on wall using four rivets. Each rivet is 6 mm in diameter and has an effective length of 12 mm. (GATE-ME-10)



Direct shear stress in MPa in the most heavily loaded rivet is

- (a) 4.4 (b) 8.8 (c) 17.6 (d) 35.2

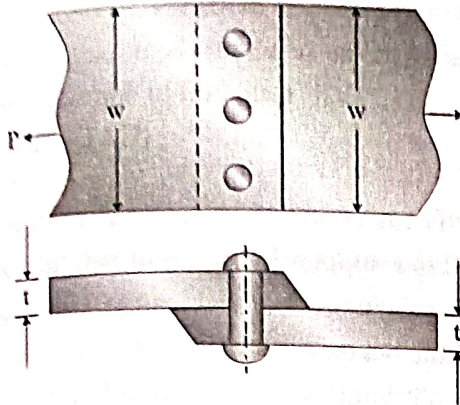
06. A fillet welded joint is subjected to transverse loading F as shown in the figure. Both legs of the fillets are of 10 mm size and the weld length is 30 mm. If the allowable shear stress of the weld is 94 MPa, considering the minimum throat area of the weld, the maximum allowable transverse load in kN is (GATE-ME-12)



- (a) 14.44 (b) 17.92
(c) 19.93 (d) 22.16

Common Data for Q.Nos. 07 & 08

A single riveted lap joint of two similar plates as shown in the figure below has the following geometrical and material details.



Width of the plate $w = 200$ mm, thickness of the plate $t = 5$ mm, number of rivets $n = 3$, diameter of the rivet $d_r = 10$ mm, diameter of the rivet hole $d_h = 11$ mm, allowable tensile stress of the plate $\sigma_p = 200$ MPa, allowable shear $\sigma_s = 100$ MPa and allowable bearing stress of the rivet $\sigma_c = 150$ MPa.

07. If the rivets are to be designed to avoid crushing failure, the maximum permissible load P in kN is (GATE-ME-13)

(a) 7.50 (b) 15.00 (c) 22.50 (d) 30.00

08. If the plates are to be designed to avoid tearing failure, the maximum permissible load P in kN is (GATE-ME-13)

(a) 83 (b) 125 (c) 167 (d) 50

09. For the three bolt system shown in the figure, the bolt material has shear yield strength of 200 MPa. For a factor of safety of 2, the minimum metric specification required for the bolt is (GATE-14-SET2)

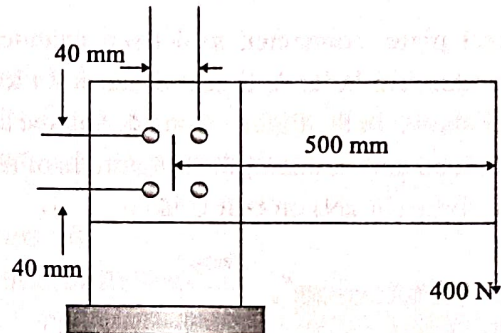


(a) M8 (b) M10
(c) M12 (d) M16

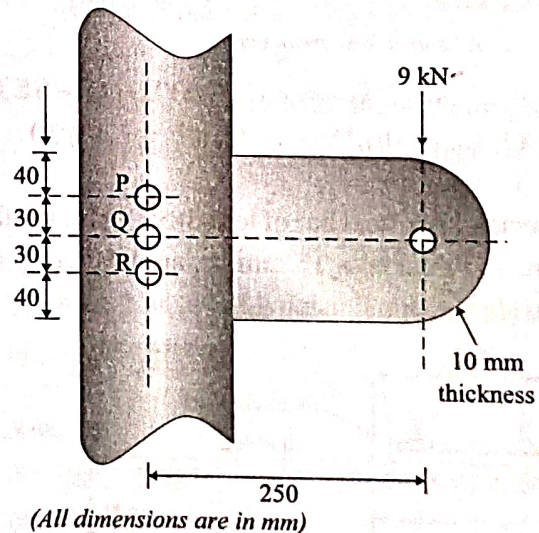
10. A bolt of major diameter 12 mm is required to clamp two steel plates. Cross sectional area of the threaded portion of the bolt is 84.3 mm^2 . The length of the threaded portion in grip is 30 mm, while the length of the unthreaded portion in grip is 8 mm. Young's

modulus of material is 200 GPa. The effective stiffness (in MN/m) of the bolt in the clamped zone is _____ (GATE-14-SET-4)

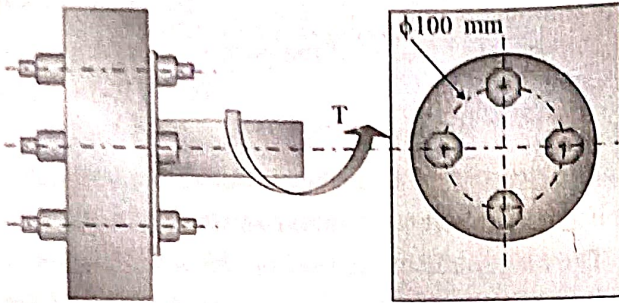
11. A horizontal plate has been joined to a vertical post using four rivets arranged as shown in the figure. The magnitude of the load on the worst loaded rivet (in N) is _____ (GATE -15 -Set 1)



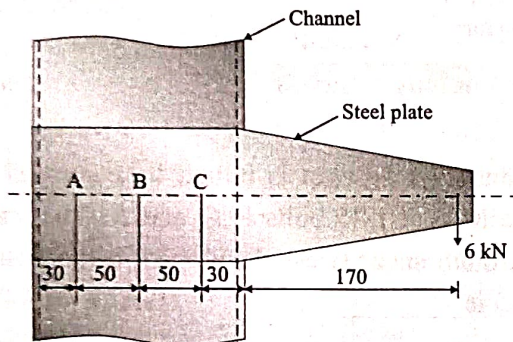
12. A cantilever bracket is bolted to a column using three M12 $\times$ 1.75 bolts P, Q and R. The value of maximum shear stress developed in the bolt P (in MPa) is _____ (GATE -15 -Set 3)



13. A bolted joint has four bolts arranged as shown in figure. The cross sectional area of each bolt is 25 mm^2 . A torque $T = 200 \text{ N-m}$ is acting on the joint. Neglecting friction due to clamping force, maximum shear stress in a bolt is _____ MPa. (GATE - 16 - SET - 3)



14. A steel plate, connected to a fixed channel using three identical bolts A, B and C, carries a load of 6 kN as shown in the figure. Considering the effect of direct load and moment, the magnitude of resultant shear force (in kN) on bolt C is

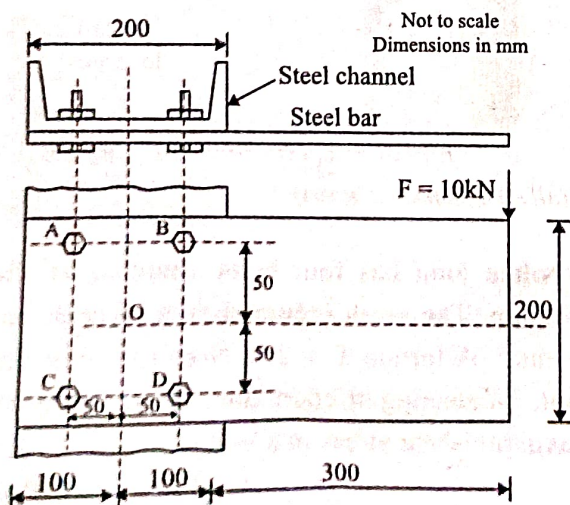


All dimensions are in mm

(GATE - 17 - SET - 2)

- (a) 13 (b) 15 (c) 17 (d) 30

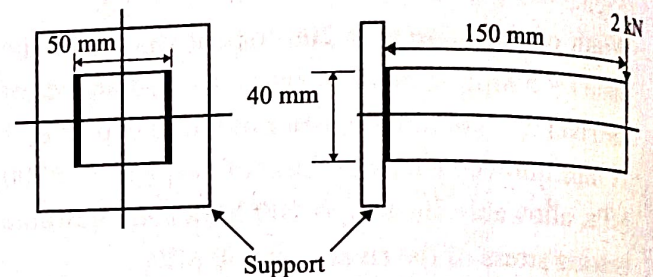
15. A rectangular steel bar of length 500 mm, width 100 mm, and thickness 15 mm is cantilevered to a 200 mm steel channel using 4 bolts, as shown.



Not to scale
Dimensions in mm

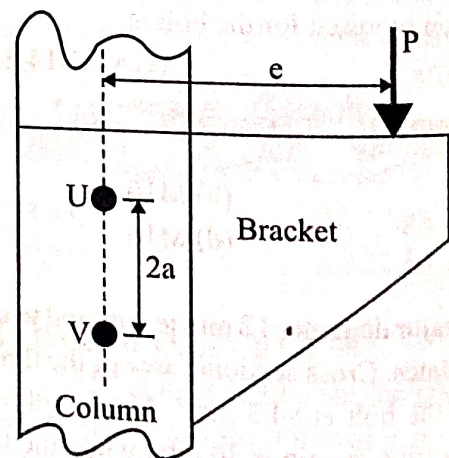
For an external load of 10 kN applied at the tip of the steel bar, the resultant shear load on the bolt at B, is _____ kN (round off to one decimal place).
(GATE-20-SET-1)

16. A cantilever beam of rectangular cross-section is welded to a support by means of two fillet welds as shown in figure. A vertical load of 2 kN acts at free end of the beam.



Considering that the allowable shear stress in weld is 60 N/mm², the minimum size (leg) of the weld required is _____ mm (round off to one decimal place).
(GATE-21-SET-1)

17. A bracket is attached to a vertical column by means of two identical rivets U and V separated by a distance of $2a = 100$ mm, as shown in the figure. The permissible shear stress of the rivet material is 50 MPa. If a load $P = 10$ kN is applied at an eccentricity $e = 3\sqrt{7}a$, the minimum cross-sectional area of each of the rivets to avoid failure is _____ mm².



(GATE-22-SET-1)

- (a) 800 (b) 25
(c) $100\sqrt{7}$ (d) 200

A key of width and height of 6 mm each is used to fix a gear on a shaft of 20 mm diameter. The shaft is used to transmit 10 kW power at 600 rpm to the gear. Permissible shear stress in the key is 80 N/mm^2 , while compressive stress in the key is neglected. The minimum length of the key, in mm, is ____ (round off to 2 decimal places).

(GATE_PI-23)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (c)	02. (b)	03. (a)	04. (c)	05. (c)
06. (c)	07. (c)	08. 2100		

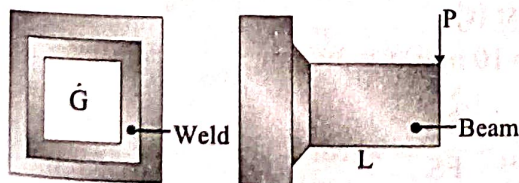
TWO MARKS QUESTIONS

01. (b)	02. (a)	03. (a)	04. (b)	05. (b)
06. (c)	07. (c)	08. (c)	09. (b)	
10. 468.827 MN/m	11. 1835 to 1845			
12. 332 to 494	13. 40	14. (c)	15. 16	
16. 6.7	17. (a)	18. 33.16		

One Mark Solutions

01. Ans: (c)

Sol:



A beam is welded as shown to a plate. It is subjected to load 'P'. 'G' is the centre of gravity of the weld area.

If we shift the load P to centre of gravity G, it will have two effects.

1. Transverse shearing due to load P.
 2. Bending of weld due to bending moment $P \times L$.
- Hence welds are subjected to combination of bending and shear.

02. Ans: (b)

$$\text{Sol: } \eta = \frac{p-d}{p} = 1 - \frac{d}{p} = 1 - 0.25 = 0.75$$

03. Ans: (a)

Sol: The bolts in rigid coupling are subjected to transverse shear and bending.

04. Ans: (c)

Sol: Effect of Pretension:

- Pre-tensioning helps to seal/prevents the leakage from the pressure vessel.
- Pre-tensioning increases fatigue life of bolts
- Pre-tensioning increases the tensile stress in the bolt.

$$F_b = F_i + \left(\frac{k_b}{k_b + k_m} \right) P$$

$$\Rightarrow F_b = F_i + CP$$

- Pre-tensioning reduces the stress amplitude i.e., decreases the deleterious effect of pressure pulsations.

05. Ans: (c)

Sol: The weld represented by the dark region in the figure is a fillet weld.

06. Ans: (c)

Sol: Pre-tensioning of a bolted joint increases stiffness of bolted joint.

07. Ans: (c)

Sol: The rivet is under double shear

$$\tau = \frac{F}{2A} = \frac{F}{2 \times \frac{\pi}{4} \times d^2} = \frac{2F}{\pi d^2}$$

$$\therefore d = \sqrt{\frac{2F}{\pi \cdot \tau}}$$

08. Ans: (2100)

Sol: Given data:

M30 bolt $\Rightarrow d = 30 \text{ mm}$

Initial tightening Load, $F_i = 350 \text{ kN}$

Torque coefficient, $K = 0.2$

$$\begin{aligned}\text{Torque required for pretension} &= K \times F_i \times d \\ &= 0.2 \times 350 \times 10^3 \times 30 \times 10^{-3} \\ &= 2100 \text{ Nm}\end{aligned}$$

Two Marks Solutions

01. Ans: (b)

Sol: $P = 0.707 S \ell \left(\frac{S_{sy}}{FS} \right)$

$$15 \times 10^3 \text{ N} = 0.707 \times 6 \times 60 \times \frac{(200)}{FS}$$

$$FS = 3.39$$

02. Ans: (a)

Sol: Primary shear force,

$$F_s = \frac{10}{3} = 3.33 \text{ kN} = F_{sB}$$

Secondary shear force, $F_s^1 = \frac{P \times e}{\sum_{i=1}^3 r_j^2} \times r$

$$F_{sA}^1 = \frac{10 \times 150}{40^2 + 0^2 + 40^2} \times 40 = 18.75 \text{ kN}$$

$$F_{sB}^1 = 0 \quad (\because r_B = 0)$$

(Centroid of bolt group lies at B)

Resultant force on A,

$$F = \sqrt{3.33^2 + 18.75^2} = 19.04 \text{ kN}$$

Stress on A,

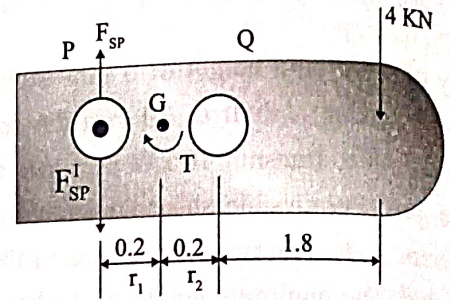
$$\sigma_A = \frac{F}{\frac{\pi}{4} \times 10^2} = \frac{19.04 \times 10^3 \text{ N}}{\frac{\pi}{4} \times 10^2} = 242.5 \text{ N/mm}^2$$

Stress on B

$$\sigma_B = \frac{3.33 \times 10^3 \text{ N}}{\frac{\pi}{4} \times 10^2} = 42.4 \text{ N/mm}^2$$

03. Ans: (a)

Sol:



Primary shear force

$$F_{SP} = F_{SQ} = \frac{4}{2} = 2 \text{ kN}$$

Secondary shear force

$$\begin{aligned}F_{SP}^1 &= \frac{P \times e}{r_1^2 + r_2^2} \times r_1 \\ &= \frac{4 \times 2}{0.2^2 + 0.2^2} \times 0.2 = 20 \text{ kN}\end{aligned}$$

04. Ans: (b)

Sol: $F_s = F_{SP}^1 - F_{SP} = 20 - 2 = 18 \text{ kN}$

$$\sigma = \frac{18 \times 10^3}{\frac{\pi}{4} \times 12^2} = 159 \text{ MPa}$$

05. Ans: (b)

Sol: Direct shear stress in all the results is same

$$F_s = \frac{100}{4} = 250 \text{ N}$$

$$\sigma = \frac{250}{\frac{\pi}{4}(6)^2} = 8.84 \text{ N/mm}^2$$

06. Ans: (c)

Sol: $S = 10 \text{ mm}$, $\ell = 30 \text{ mm}$,

$$\tau_{\text{weld}} = \frac{S_{sy}}{FS} = 94 \text{ MPa}$$

Maximum allowable transverse load, (in kN),

$$\begin{aligned}F &= 0.707 S \times \ell \times \tau_{\text{weld}} \\ &= 0.707 S \times \ell \times \frac{S_{sy}}{FS} \\ &= 0.707 \times 10 \times 30 \times 94 = 19.93 \text{ kN}\end{aligned}$$

07. Ans: (c)

$$\begin{aligned}\text{Sol: } P &= d_r \times t \times \sigma_c \times n \\ &= 10 \times 5 \times 150 \times 3 \\ &= 22500 \text{ N} \\ &= 22.5 \text{ kN}\end{aligned}$$

08. Ans: (c)

$$\begin{aligned}\text{Sol: } P &= (W - 3d_h)t \times \sigma_p \\ &= (200 - 3 \times 11) \times 5 \times 200 \\ &= 167 \times 5 \times 200 = 167 \text{ kN}\end{aligned}$$

09. Ans: (b)

$$\text{Sol: } S_{sy} = 200 \text{ MPa, F.S} = 2$$

$$\tau = \frac{\text{Load}}{\text{No. of bolts} \times \text{area}} = \frac{19 \times 10^3}{3 \times \frac{\pi}{4} d^2}$$

$$\Rightarrow \frac{200}{2} = \frac{19 \times 10^3}{3 \times \frac{\pi}{4} \times d^2}$$

$$d = 8.97 \approx 9$$

$$\therefore d = \text{M10}$$

10. Ans: 468.827

$$\text{Sol: } d = 12 \text{ mm, } E = 2 \times 10^5 \text{ N/mm}^2$$

$$A_1 = 84.3 \text{ mm}^2, A_2 = \frac{\pi}{4} \times 12^2 \text{ mm}^2$$

$$\ell_1 = 30 \text{ mm, } \ell = 30 + 8 = 38 \text{ mm}$$

$$\ell_2 = 8 \text{ mm, } K = ?$$

$$K = \frac{W}{\delta} = \frac{W}{\left(\frac{W\ell}{AE}\right)} = \frac{AE}{\ell}$$

$$\delta\ell_1 = \frac{W\ell_1}{A_1 E_1} = \frac{W \times 30}{84.3 \times 2 \times 10^5} \text{ mm}$$

$$\delta\ell_2 = \frac{W\ell_2}{A_2 E_2} = \frac{W \times 8}{\frac{\pi}{4} \times 12^2 \times 2 \times 10^5}$$

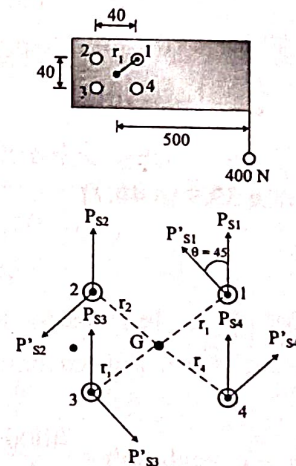
$$= \frac{W \times 8}{113 \times 2 \times 10^5} \text{ mm}$$

$$K = \frac{W}{\delta\ell_1 + \delta\ell_2} = \frac{W}{\left(\frac{W}{2 \times 10^5}\right) \left[\frac{30}{84.3} + \frac{8}{113}\right]}$$

$$\begin{aligned}&= \frac{2 \times 10^5}{0.3558 + 0.07079} = \frac{2 \times 10^5}{0.4265} \\ &= 468.827 \text{ MN/m}\end{aligned}$$

11. Ans: 1835 to 1845

Sol: Worst loaded rivet is 1 or 4



$$\begin{aligned}r_1 = r_2 = r_3 = r_4 &= \sqrt{20^2 + 20^2} \\ &= \sqrt{2} \times 20 = 28.28 \text{ mm}\end{aligned}$$

Primary shear load,

$$P_{s1} = \frac{400}{4} = 100 \text{ N} = P_{s2} = P_{s3} = P_{s4}$$

Secondary shear load,

$$\begin{aligned}P'_{s1} &= \frac{P \times \ell}{r_1^2 + r_2^2 + r_3^2 + r_4^2} \times r_1 \\ &= \frac{400 \times 500}{4 \times 28.28^2} \times 28.28 = 1768 \text{ N}\end{aligned}$$

$$\begin{aligned}P_{\text{resultant}} &= \sqrt{P_{s1}^2 + P_{s1}^2 + 2P_{s1}P'_{s1} \cos \theta} \\ &= \sqrt{100^2 + 1768^2 + 2 \times 100 \times 1768 \times \cos 45} \\ &= 1840 \text{ N}\end{aligned}$$

12. Ans: 332 to 494

Sol: Primary load on 'P'

$$P_{s1} = \frac{9}{3} = 3 \text{ kN}$$

Secondary load on 'P'

$$P'_{s1} = \frac{P \times \ell}{r_1^2 + r_2^2 + r_3^2} \times r_1$$

$$= \frac{9 \times 250}{30^2 + 0 + 30^2} \times 30$$

$$= \frac{9 \times 250}{2 \times 30^2} \times 30 = 37.5 \text{ kN}$$

Resultant load,

$$P_s = \sqrt{P_s^2 + P'_s{}^2} = \sqrt{3^2 + 37.5^2} = 37.61 \text{ kN}$$

$$\tau = \frac{P_s}{\frac{\pi}{4} d^2} = \frac{37.61 \times 10^3}{\frac{\pi}{4} \times 12^2} = 332.63 \text{ MPa}$$

13. Ans: 40 (range 39.9 to 40.1)

Sol: $T = F \times \frac{D}{2}$

$$200 = F \times \frac{0.1}{2}$$

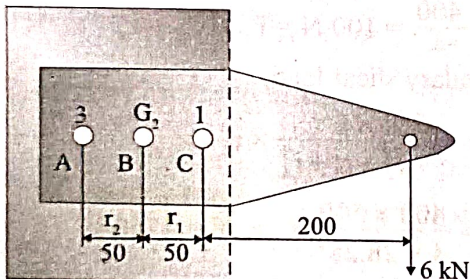
$$F = 4000 \text{ N}$$

$$\Rightarrow \text{Shear force in each bolt} = \frac{4000}{4} = 1000$$

$$\Rightarrow \tau = \frac{1000}{25} = 40 \text{ MPa}$$

14. Ans: (c)

Sol:



Primary shear, load, $P_{s1} = \frac{6}{3} = 2 \text{ kN}$

Secondary shear, load, $P'_{s1} = kr_1$

$$= \frac{P \times e}{r_1^2 + r_2^2 + r_3^2} \times r_1$$

$$r_1 = r_3 = 50 \text{ \& } r_2 = 0 \text{ \& } e = 250 \text{ mm}$$

$$\therefore P'_{s1} = \frac{6 \times 250}{50^2 + 0^2 + 50^2} \times 50$$

$$= \frac{6 \times 250}{2 \times 50^2} \times 56 = 15 \text{ kN}$$

$\therefore$ Resultant load on c,

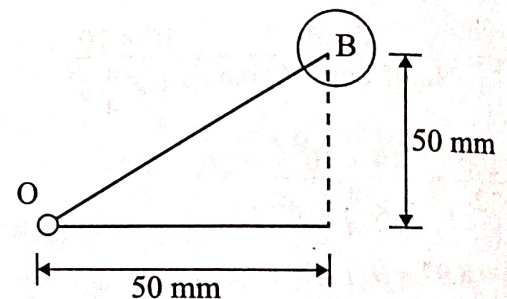
$$P_s = P_{s1} + P'_{s1} = 2 + 15 = 17 \text{ kN}$$

15. Ans: 16

Sol: Due to symmetry,

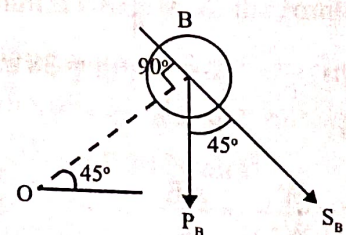
Primary load on B (P_B) = $\frac{10}{4} = 2.5 \text{ kN}$

Secondary load on B (S_B) = $\frac{F \cdot e}{r_1^2 + r_2^2 + r_3^2 + r_4^2} \times r$
 $= \frac{10 \times 400}{4(50\sqrt{2})^2} \times 50\sqrt{2}$
 $= 10\sqrt{2} \text{ kN}$



$$OB = r = r_1 = r_2 = r_3 = r_4 = \sqrt{50^2 + 50^2} = 50\sqrt{2}$$

Free Body Diagram of B



Here the angle between P_B and S_B is 45° .

$$\therefore \theta = 45^\circ$$

Resultant shear load acting on B

$$R_B = \sqrt{P_B^2 + S_B^2 + 2P_B \cdot S_B \cdot \cos \theta}$$

$$= \sqrt{2.5^2 + (10\sqrt{2})^2 + 2 \times 2.5 \times 10\sqrt{2} \cdot \cos 45^\circ}$$

$$R_B = 16 \text{ kN}$$

16. Ans: (6.7)

Sol:

(i) Direct shear stress,

$$\tau = \frac{P}{A} = \frac{2 \times 10^3}{2 \times t \times 40} = \frac{25}{t} \text{ N/mm}^2$$

(ii) Bending stress (σ_b)

$$\sigma_b = \frac{M y}{I} = \frac{2000 \times 150 \times \frac{40}{2}}{2 \times \frac{t \times 40^3}{12}}$$

$$\sigma_b = \frac{562.5}{t} \text{ N/mm}^2$$

(iii) Using shear stress theory :

$$\left\{ \left(\frac{\sigma_b}{2} \right)^2 + \tau^2 \right\}^{0.5} \leq \tau_{\text{allow}}$$

$$\left\{ \left(\frac{562.5}{2 \times t} \right)^2 + \left(\frac{25}{t} \right)^2 \right\}^{0.5} = 60$$

$$\therefore t = 4.7 \text{ mm}$$

$$\text{Weld size, } h = t \times \sqrt{2} = 6.66 \text{ mm} = 6.7 \text{ mm}$$

17. Ans: (a)

Sol: The given load is eccentric lateral load which results in

(i) primary shear due to direct loading

(ii) secondary shear due to eccentricity

(i) Primary shear:

$$F_p = \frac{F}{n} = \frac{10 \text{ kN}}{2} = 5 \text{ kN}$$

(ii) Secondary shear:

$$F_s = \frac{P \times e}{r_1^2 + r_2^2} \times r = \frac{10 \times 3\sqrt{7}a}{a^2 + a^2} \times a \quad (a = 50 \text{ mm})$$

$$= 15\sqrt{7} \text{ kN}$$

Finding resultant:

$$R = \sqrt{F_p^2 + F_s^2 + 2F_p F_s \cos \theta}$$

Here θ is 90°

As secondary load is same on both rivets. Both are critical due to loading.

$$\therefore R_{\text{max}} = \sqrt{F_p^2 + F_s^2}$$

$$= \sqrt{(5)^2 + (15\sqrt{7})^2}$$

$$R_{\text{max}} = 40 \text{ kN}$$

Design of Rivet:

$$\tau_{\text{max}} = \frac{S_{\text{ys}}}{\text{FOS}}$$

$$\frac{R_{\text{max}}}{A} = \frac{50}{1}$$

$$\frac{40 \times 10^3}{A} = 50$$

$$A = 800 \text{ mm}^2$$

As FOS is considered as 1, A represents the minimum cross section area required.

18. Ans: 33.16

Sol: Torque transmitted,

$$T = \frac{60P}{2\pi N} = \frac{60 \times 10 \times 10^3}{2\pi \times 600} = 159.15 \text{ N-m}$$

Tangential load (or) shear load on key,

$$F = \frac{T}{r} = \frac{159.15}{0.01} = 15.915 \text{ kN}$$

To avoid failure, $F \leq F_s$

$$F \leq S_{\text{ys}} \cdot A_s$$

$$A_s = \text{Shear area} = l \times b$$

l = length of key

b = width of key

$$\therefore 15.915 \times 10^3 \leq 80 \times l \times 6$$

$$\therefore l \geq 33.16 \text{ mm}$$

One Mark Questions

01. For full depth of involute spur gears, minimum number of teeth of pinion to avoid interference depends upon. (GATE-ME-88)
 (a) pressure angle (b) speed ratio
 (c) circular pitch (d) pitch diameter
02. Modern gear tooth profile is given involute shape because (GATE-ME-89)
 (a) this is a very easy curve for manufacturing
 (b) sliding does not takes place anywhere on meshing teeth
 (c) involute is the only profile that gives conjugate action
 (d) change in the centre distance does not change gear ratio
03. For a pinion of 15 teeth, under cutting _____ (increases/decreases) with _____ (increase/decrease) of pressure angle. (GATE-ME-91)
04. Interference in a pair of gears is avoided, if the addendum circles of both the gears intersect common tangent to the base circles within the points of tangency. (GATE-ME-95)
 (a) True (b) False
05. Match the following: (GATE-ME-95)
List-I (Gear types)
 A. Worm gears
 B. Cross helical gears
 C. Bevel gears
 D. Spur gears
List-II (Applications)
 1. Parallel shafts
 2. Non-parallel, intersecting shafts
 3. Non-parallel, non-intersecting shafts
 4. Large speed ratios
06. In spur gears, the circle on which the involute is generated is called the (GATE-ME-96)
 (a) pitch circle (b) clearance angle
 (c) base circle (d) addendum circle
07. To make a worm drive reversible, it is necessary to increase (GATE-ME-97)
 (a) centre distance (b) worm diameter factor
 (c) number of starts (d) reduction ratio
08. A 1.5 kW motor is running at 1440 rev/min. It is to be connected to a stirrer running at 36 rev/min. The gearing arrangement suitable for this application is (GATE-ME-00)
 (a) differential gear (b) helical gear
 (c) spur gear (d) worm gear
09. Large speed reductions (greater than 20) in one stage of a gear train are possible through (GATE-ME-02)
 (a) Spur gearing (b) Worm gearing
 (c) Bevel gearing (d) Helical gearing
10. The minimum number of teeth on the pinion to operate without interference in standard full height involute teeth gear mechanism with 20° pressure angle is (GATE-ME-02)
 (a) 14 (b) 12 (c) 18 (d) 32
11. Two mating spur gears have 40 and 120 teeth respectively. The pinion rotates at 1200 rpm and transmits a torque of 20 N.m. The torque transmitted by gear is (GATE-ME-04)
 (a) 6.6 Nm (b) 20 Nm (c) 40 Nm (d) 60 Nm
12. Tooth interference in an external involute spur gear pair can be reduced by (GATE-ME-10)
 (a) decreasing center distance between gears
 (b) decreasing module
 (c) decreasing pressure angle
 (d) increasing number of teeth